Poonam Sahoo

poonamsahoo@stanford.edu | 978-460-9573 | Stanford, CA 94305

EDUCATION

Stanford University, *Stanford, CA*
M.S. in Computer Science, Artificial Intelligence Specialization. GPA: 4.04 — Expected June 2026
Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Convex Optimization

B.S. in Mathematics. GPA: 3.92 — June 2025
Relevant Coursework: Statistical Inference, Algorithms, Probability Theory
Organizations: Resident Student Leader of Okada House, Rewriting the Code, Out in STEM, South Asian Society

SKILLS

Python (Pandas, NumPy, PyTorch), R, SQL, Javascript, Typescript, Svelte, C/C++, QGIS/arcGIS

RELEVANT EXPERIENCE

Reddit, Data Science Intern (Ads Data Science), *San Francisco, CA* — June 2025 - September 2025
- Used SQL and Python to build a framework for modeling user lifetime value of logged-in Reddit users
- Collaborated with cross-functional stakeholders (Consumer, Ads Engineering) to define ambiguities and remain aligned on model goals and output
- Implemented a Markov Chain modeling approach to causally determine the long-term impacts of A/B testing experiments on Reddit ads revenue
- Backtesting on an ad-load experiment showed the Markov Chain model predicted future user behavior (churn, retention, etc.) with a 40-60% reduction in error metrics (RMSE, Cross Entropy Loss) compared to the baseline for a 90-day prediction period

Goldman Sachs, Quant Strats Software Engineering Intern, *New York, NY* — June 2024 - August 2024
- Interface with traders to build quantitative trading strategies and automate trader workflow in Python and Slang (Goldman's proprietary language)
- Recalibrate part of credit risk model for liquidation of government bonds, reducing fees for 60% of hedge fund clients while still covering 99.7th percentile risk events
- Build backtesting system for determining which day US-based indices reinvest their dividends

Stanford Human-Computer Interaction Group, Undergraduate Researcher, *Stanford, CA* — March 2022 - Present
- Led project on a communication tool to help speakers with audience understanding using steerable live comment clustering. Built a webapp with a SvelteKit frontend and Flask backend and piloted tool to 100+ student audience during CS 347 lectures. Work is accepted as a demo abstract for UIST 2025 (first-author project)
- Analyzed 470,000 online advertisements and user data with Python and SQL and created visualizations of the data to aid in research

Belvedere Trading, Junior Quantitative Trading Analyst Intern, *Chicago, IL* — June 2023 - August 2023
- Analyzed tick by tick market data to determine optimal hedge execution strategies for Tesla, Meta, Amazon, and Apple stocks while minimizing risk and maximizing PnL
- Proposed hedge execution strategy with estimated daily PnL of $10,000 and high risk-adjusted returns relative to naive benchmark
- Constructed trading infrastructure in Python for backtesting and keeping track of past trade decisions and risk

PROJECTS

U.S. Newspaper Polarization via Heterogenous GNNs | *Python, PyG, BeautifulSoup* — September 2024 - December 2024
- Supplemented a newswire dataset with webscraped election results using BeautifulSoup and geocoder API
- Training a heterogenous GNN consisting of newspaper articles and newspaper outlets to form clusters of similar newspaper outlets. Goal is to observe political polarization across decades (e.g. 1920s vs. 1960s)

Dress-UP: A Deep, Unique, Personalized Fashion Recommender | *Python, PyTorch* — April 2024 - June 2024
- Computer vision project – created a clothing recommender that builds off of CLIP and Fashion-CLIP by modifying the architecture (adding Spatial Transformer to Fashion-CLIP architecture)
- Finetuned FCLIP model and STN model beat baseline (CLIP) with regards to fashion recommendations for 20 participants with a MAP@4 of 0.56 (baseline was 0.3)